

AUTOMOBILE INSURANCE ANALYSIS

Power BI Dashboard Analysis

AUTOMOBILE INSURANCE

Content:

- **Objective:** Analyze automobile insurance performance through premiums, complaint ratios, and company-level comparisons.
- **Data Source:** Automobile Insurance dataset (2009–2016)
- **Tools Used:** Power BI, DAX Measures, Data Modelling, and Visualization

Key Focus Areas:

- Complaint diagnostics and trends
- Premium growth and performance
- Company-wise comparison and ranking.

COMPLAINS ANALYSIS AND KEY TRENDS

Content:

- **Total Complaints:** 35.9K
- **Average Complaint Ratio:** 1.89%
- **YoY Complaint Change %:**   10.8% (Increase)
- **Top 5 Companies by Complaint Ratio:**
 - Clarendon National Insurance Company
 - OneBeacon Insurance Company
 - Greater New York Mutual Insurance Company
 - Zurich American Insurance Company
 - National Continental Insurance Company

Key Visuals (from dashboard):

- “Complaint Type Distribution by Year” chart
- “Complaint Type Composition” pie chart (Upheld / Not Upheld / Question of Fact)

Insight Bullets:

- Majority (56%) of complaints were *Not Upheld*.
- Steady rise in complaints observed from 2012 to 2016.
- Complaint ratios higher for smaller-scale companies.

Complaint Diagnostic
(Company-focused)

Total Companies

1358

Total Complaints

35.9K

Average complaint ratio %

1.89%

YoY Complaint Change %

10.80%

View Premium Trends
on next slide



Complaint Types

All

Filing_Year

Select all

2011

2014

2009

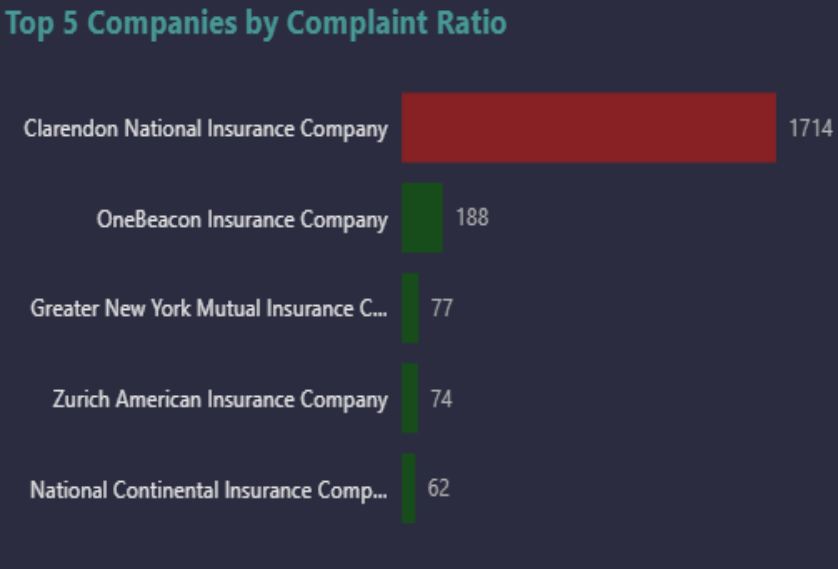
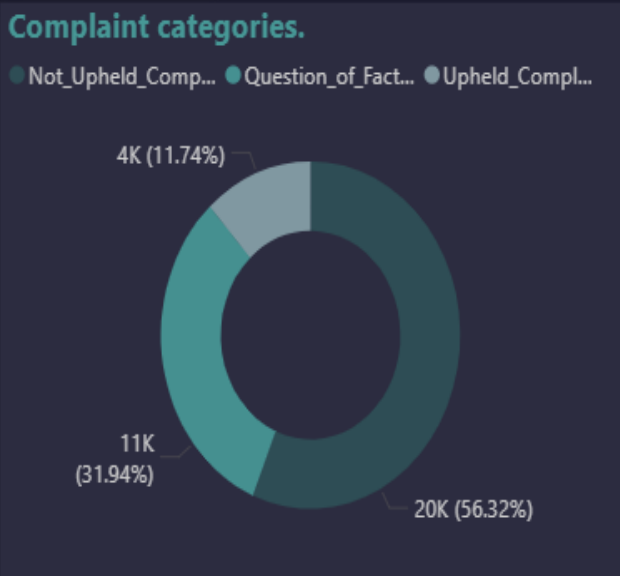
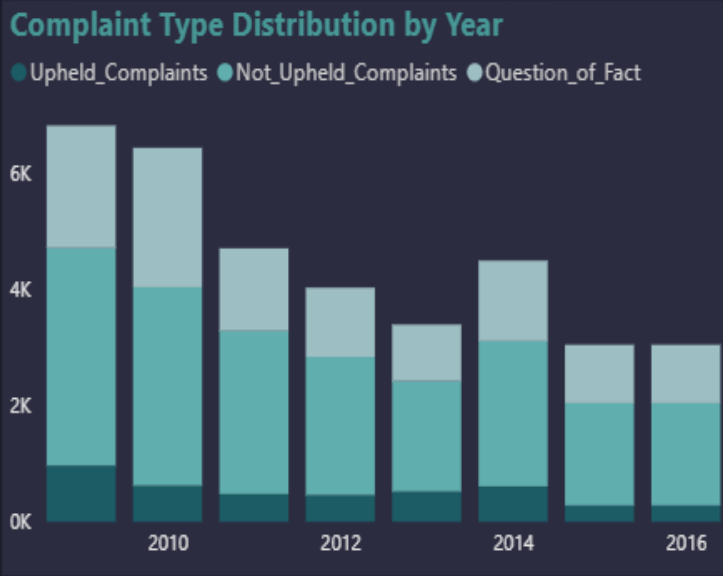
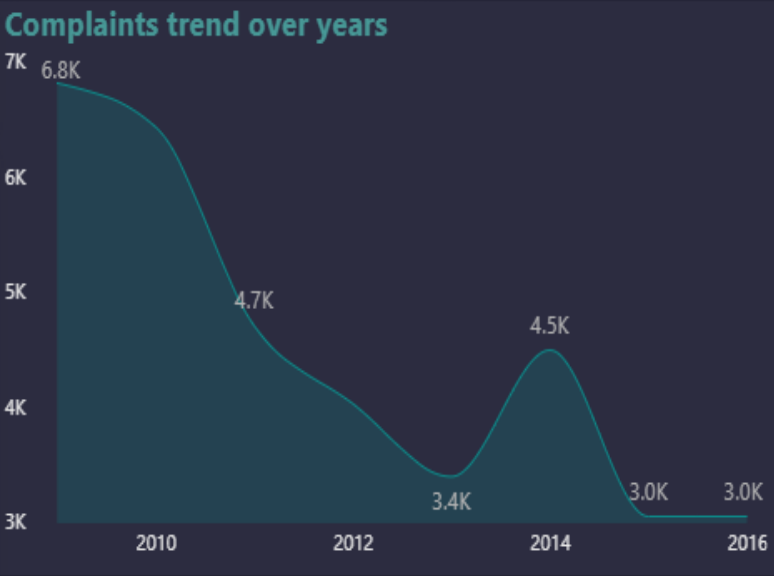
2012

2015

2010

2013

2016



Company_Name	Total Complaints	Upheld	Not Upheld	Question_of_Fact
Foremost Signature Insurance Company	8	0	6	2
Foremost Insurance Company Grand Rapids, Michigan	67	3	47	17
First Liberty Insurance Corporation	68	10	31	27
Fireman's Fund Insurance Company	6	1	4	1
Federal Insurance Company	16	2	8	6
Farmington Casualty Company	69	6	47	16
Farmers New Century Insurance Company	68	13	31	24
Farm Family Casualty Insurance Company	68	11	39	18
Eveready Insurance Company	143	8	103	32
Esurance Property and Casualty Insurance	36	1	27	8
Esurance Insurance Company	283	38	190	55
Essentia Insurance Company	7	1	5	1
Total	35919	4216	20229	11474

PREMIUM PERFORMANCE AND BUSINESS GROWTH

Content:

- Total Premiums Written: 87.43K
- YoY Premium Growth:   6.12%
- Average Premium per Company: 64.38%
- Insight Highlights:
 - Premiums increased steadily across years with slight dips in 2012–2014.
 - Growth in premiums correlated with improvement in complaint ratios (quality control).
 - Premium vs Complaint Ratio chart shows positive relationship at moderate scales.

Key Visuals (from dashboard):

- “Premiums Written over Filing Year” (Line Chart)
- “Premiums vs Complaint Ratio Over Time” (Combo Chart)

Conclusion:

The market shows sustainable growth and improving complaint management with increased business scale.

Premium & Trend Analysis

Total Premiums Written

87.43K

Average Premium Per Company %

64.38%

YoY Premium Growth %

6.12%

[View Company Comparison on next slide](#)



Complaint Types

All

Filing Year

Select all

2011

2014

2009

2012

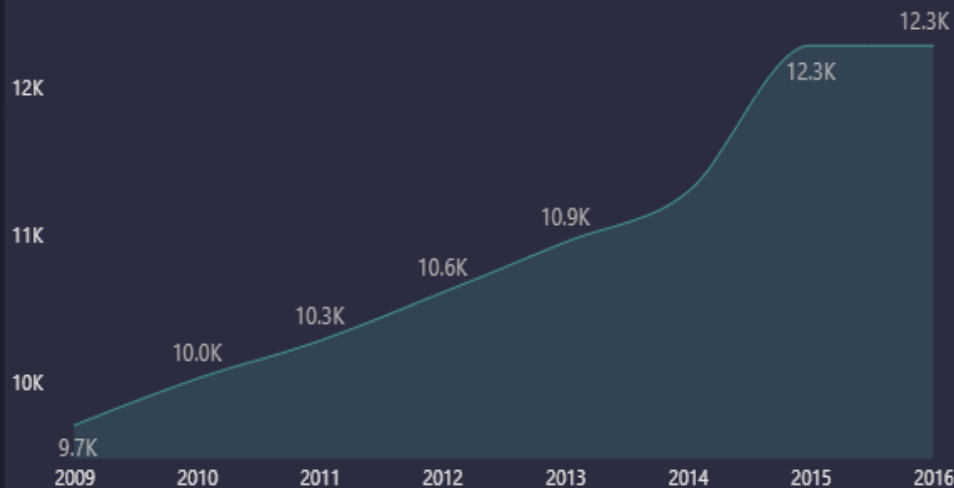
2015

2010

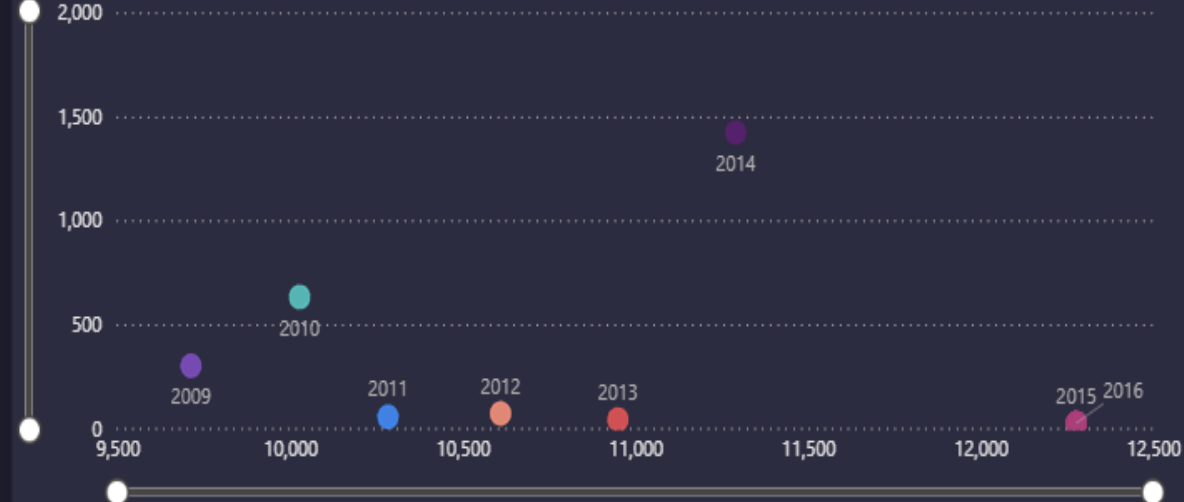
2013

2016

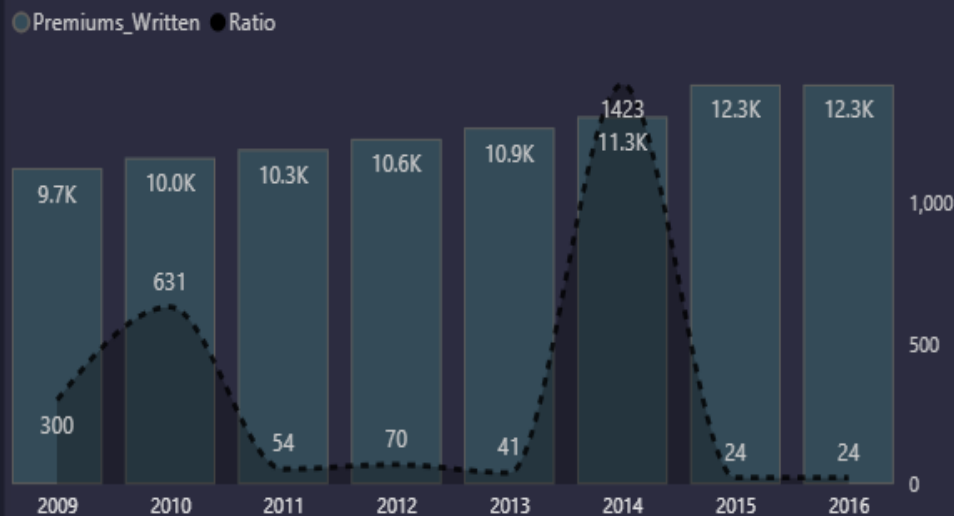
Premiums Written over Filing Year



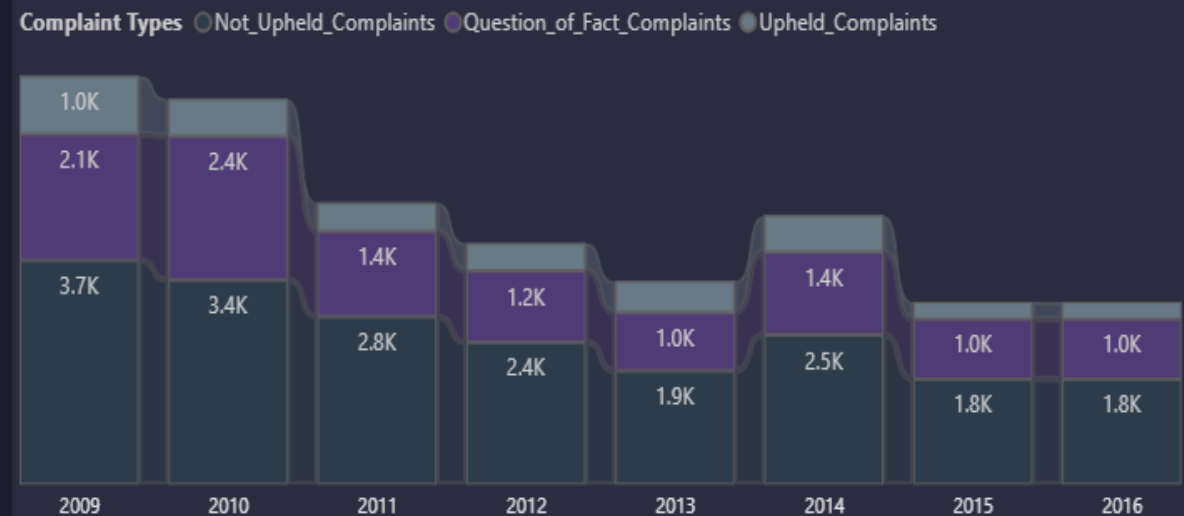
Complaint Ratio vs Premiums Written



Premiums Written vs Complaint Ratio Over Time



Complaint Resolution Trends vs Business Scale



COMPANY-WISE PERFORMANCE COMPARISON

Content:

- Top Performing Company : Progressive Specialty Insurance Company
- Bottom Performing Company : New Hampshire Insurance Company
- Highest Premiums Written : GEICO General Insurance Company
- Average Complaint Ratio (Top 10) : 0.41%

Key Visuals:

- “Composite Performance Rank”
- “YoY Premium & Complaint Ratio Trend”
- “Complaint Resolution Rate by Company”

Insights:

- GEICO leads in premium volume but shows moderate complaint resolution (28.8%).
- Progressive shows balanced growth and lowest complaint ratio → top rank.
- Companies with lower premium scale often show weaker complaint handling.

Company Comparison Dashboard

Complaint Types

All

Filing_Year

Select all

2011

2014

2009

2012

2015

2010

2013

2016

Top Performing Company

Progressive Specialty Insurance Company

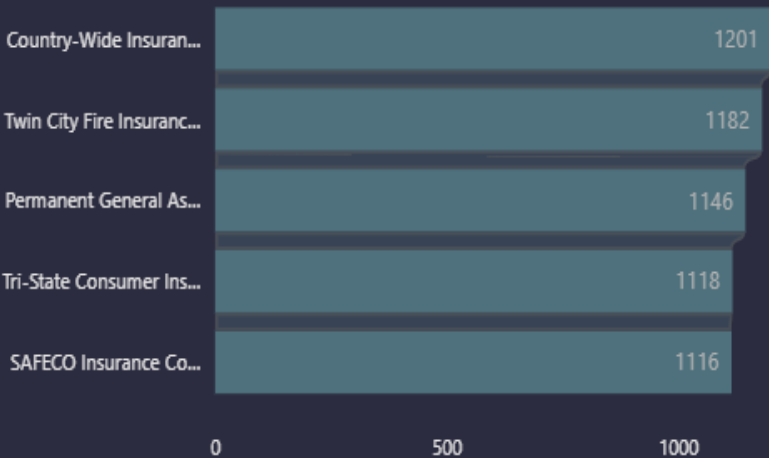
Bottom Performing Company

New Hampshire Insurance Company

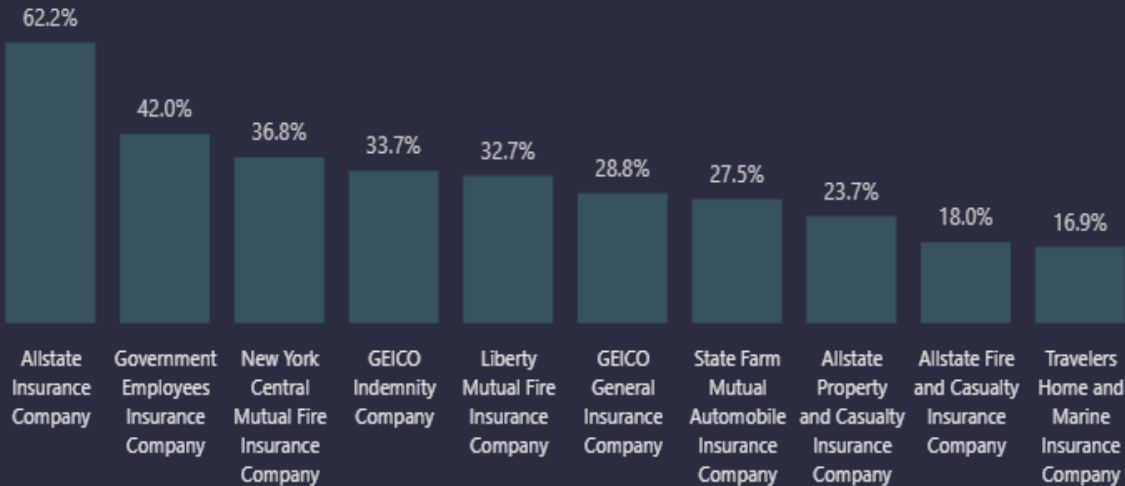
Highest Premiums Written

GEICO General Insurance Company

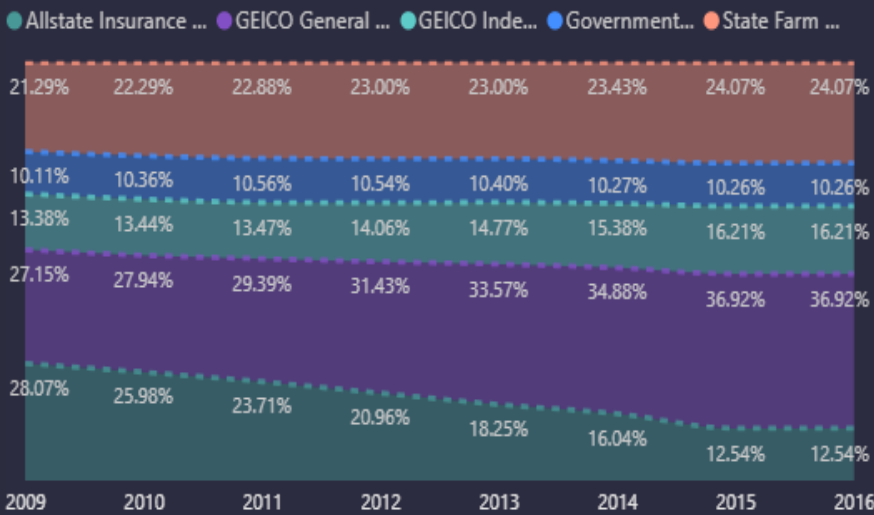
Composite Performance Rank



Complaint Resolution Rate by Company



YoY Premium & Complaint Ratio Trend



Company_Name	Total_Complaints	Complaint_Ratio	Premiums_Written	Ranking
GEICO General Insurance Company	3898	0.29	13,514.38	650
State Farm Mutual Automobile Insurance Company	2629	0.28	9,557.52	781
Allstate Insurance Company	4949	0.62	7,951.47	979
GEICO Indemnity Company	2057	0.34	6,099.02	651
Government Employees Insurance Company	1797	0.42	4,281.05	658
Allstate Property and Casualty Insurance Company	923	0.24	3,887.04	840
Liberty Mutual Fire Insurance Company	1113	0.33	3,405.32	794
New York Central Mutual Fire Insurance Company	725	0.37	1,971.81	653
Allstate Fire and Casualty Insurance Company	343	0.18	1,910.72	484
Travelers Home and Marine Insurance Company	311	0.17	1,837.75	795
Progressive Casualty Insurance Company	682	0.39	1,748.30	857
Total	35919	0.41	87,427.28	116043

STRATEGIC TAKEAWAYS

Insights:

- Complaint ratios rising **10.8% YoY** – needs targeted service quality improvement.
- Strong premium growth (+**6.12%**) shows market expansion.
- High complaint resolution rate companies ($\geq 50\%$) show stronger brand trust.

Recommendations:

- Strengthen customer service in low-resolution companies.
- Focus premium growth strategies on mid-performing companies with stable ratios.
- Regularly monitor complaint ratio and resolution KPIs to sustain trust.
- Continue tracking YoY trends for performance stability.

Total Complaints

35.9K

YoY Complaint Change %



10.80%

Total Premiums Written

87.43K

YoY Premium Growth %



6.12%

THANK YOU...